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Experiment 2

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Subject Name: Python Programming **Subject Code: 25CAH-606**

1. **Aim/Overview of the practical:** WAP to perform list operation-creating, accessing, slicing, modifying, adding, removing, sorting, reversing, finding length, and looping through lists

2. **Algorithm/Flowchart :**

Algorithm for List Operations:

1. **Initialize Empty List:**

   - Create an empty list named List.
   - Print the text "25MCA:" and the empty list.

2. **Create and Print List of Numbers:**

   - Initialize List with numbers [10, 20, 15].
   - Print the text "List of numbers:" and the list.

3. **Create and Print List of Strings:**

   - Initialize List with strings ["BCA", "MCA", "UIC"].
   - Print the text "List Items:".
   - Print the first item (List[0]).
   - Print the third item (List[2]).

4. **Access List Elements Using Negative Indexing:**

   - Reinitialize List with mixed elements [5, 6, 'BCA', 5, 'MCA', 6, 'UIC'].
   - Print the text "Accessing element using negative indexing".
   - Print the last element (List[-1]).

- Print the third last element (List[-3]).

5. **Addition of Elements in a List:**

- Initialize an empty list List.
- Print "EMPTY LIST:" and the empty list.

6. **Add Individual Elements to List:**

- Append 1, 2, and 5 to List.
- Print "List after Addition of Three elements:" and the list.

7. **Add Multiple Elements Using a Loop:**

- Use a loop to append elements from 1 to 3 to List.
- Print "List after Addition of elements from 1-3:" and the list.

8. **Add Tuple to List:**

- Append the tuple (5, 6) to List.
- Print "List after Addition of a Tuple:" and the list.

9. **Add a List to the List:**

- Append a list ['BCA', 'MCA'] to List.
- Print "List after Addition of a List:" and the list.

Algorithm for Dictionary Operations:

1. **Initialize and Print an Empty Dictionary:**

- Create an empty dictionary Dict.
- Print "Empty Dictionary:" and the empty dictionary.

2. **Create and Print Dictionary Using dict() Constructor:**

- Initialize Dict using the dict() constructor with pairs (1: 'Ph.D', 2: 'PG', 3: 'UG').
- Print "Dictionary with the use of dict():" and the dictionary.

3. **Create and Print Dictionary Using List of Pairs:**

- Initialize Dict with a list of pairs [(1, 'Ph.D'), (2, 'PG')].
- Print "Dictionary with each item as a pair:" and the dictionary.

3. Code for experiment/practical:

```
# Creating a List List = [] print("24MCA: ") print(List)

# Creating a List of numbers List = [10, 20, 14]
print("\nList of numbers: ") print(List)

# Creating a List of strings and accessing # using index
List = ["BCA", "MCA", "UIC"]
print("\nList Items: ") print(List[0])
print(List[2])

List = [5, 6, 'BCA', 4, 'MCA', 6, 'UIC']

# accessing an element using # negative indexing
print("Accessing element using negative indexing")

# print the last element of list print(List[-1])

# print the third last element of list print(List[-3])

# Python program to demonstrate # Addition of elements in a List

# Creating a List List = [] print("EMPTY LIST: ")
print(List)
```

```

# Addition of Elements # in the List List.append(1) List.append(2) List
    .append(4)
print("\nList after Addition of Three elements: ") print(List)

# Adding elements to the List # using Iterator
for i in range(1, 4): List.append(i)
print("\nList after Addition of elements from 1-3: ") print(List)

# Adding Tuples to the List List.append((5, 6))
print("\nList after Addition of a Tuple: ") print(List)

# Addition of List to a List List2 = ['BCA', 'MCA']
List.append(List2)
print("\nList after Addition of a List: ") print(List)

```

DICTIONARY OPERATIONS

```
Dict = {}
print("Empty Dictionary: ") print(Dict)

Dict = dict({1: 'Ph.D', 2: 'PG', 3: 'UG'})
print("\nDictionary with the use of dict(): ") print(Dict)

Dict = dict([(1, 'Ph.D'), (2, 'PG')])
print("\nDictionary with each item as a pair: ") print(Dict)

```

4. Result/Output/Writing Summary:

List Operations:

```
24MCA:
[]

List of numbers:
[10, 20, 14]

List Items:
BCA
UIC
Accessing element using negative indexing
UIC
MCA
EMPTY LIST:
[]

List after Addition of Three elements:
[1, 2, 4]

List after Addition of elements from 1-3:
[1, 2, 4, 1, 2, 3]

List after Addition of a Tuple:
[1, 2, 4, 1, 2, 3, (5, 6)]

List after Addition of a List:
[1, 2, 4, 1, 2, 3, (5, 6), ['BCA', 'MCA']]
```

Dictionary Operations

```
Empty Dictionary:
{}

Dictionary with the use of dict():
{1: 'Ph.D', 2: 'PG', 3: 'UG'}

Dictionary with each item as a pair:
{1: 'Ph.D', 2: 'PG'}
```

5. Learning outcomes (What I have learnt):

1. I have learnt how to create both empty and initialized lists in Python, using different data types such as integers, strings, and mixed data types.
2. I have learnt how to use loops to append multiple elements to a list.
3. I have learnt how to create tuples and explore how to nest tuples within each other using loops.
4. I have learnt how to create an empty dictionary and initialize a dictionary with key-value pairs using the dict() constructor.